

Kindergarten Ontario Curriculum Mathematical Learning

Data Detectives

Unit Overview: Data Detectives

Grade: Kindergarten | **Strand:** Mathematical Learning

Target Expectations: A8.1, A8.2, A8.3, A8.4

Learning Goal: Children will categorise, count, display, and describe simple data.

Lesson	Lesson Title	Learning Goal
Day 1	Sorting and Categorizing	I can sort objects into groups.
Day 2	Counting Data	I can collect data by counting things in groups.
Day 3	Picture Graphs	I can make a picture graph.
Day 4	Likelihood: Always, Sometimes, Never	I can tell if events ALWAYS, SOMETIMES, or NEVER happen.
Day 5	Data Detective Capstone	I can be a Data Detective for a question I pick.

Lesson 1: Sorting and Categorizing

Goal: Things can be sorted into groups by a SAME attribute — colour, shape, size.

Expectation: A8.1

Learning Goal: "I can sort objects into groups."

Materials: M1_sort_mats, M2_button_set, M5_question_cards, M7_object_set

1. Minds On (10 mins): Detective Dot's Acorn Sort

Detective Dot has a pile of acorns! Help her SORT them by COLOUR.

Bring children to the carpet. Detective Dot puppet. 'Friends — I'm Detective Dot the Sorting Squirrel. I have a pile of acorns. Some are big, some small. Some red, some brown. I need to SORT them.'

2. Action (18 mins): Sorting Stations

- Whole group: sort 12 buttons together.
- Three stations.
- Stretch challenge at Station C: sort the same set TWO different ways.
- Discuss attributes.
- Circulate to record formative evidence on the sorting tracker.
- Two pairs share their sorts of the same set with DIFFERENT attributes (one by colour, one by size).

3. Consolidation (7 mins): Detective Dot's Sort Brag

Discuss:

- What does SORT mean?
- What attributes did you use?
- Can the same things be sorted different ways?
- What's a category?

Exit: Detective Dot's refrain — class chants 3x: 'ONE attribute — ONE sort — ONE detective!' Each child writes one finding on a sticky.

Lesson 2: Counting Data

Goal: Counting things in groups gives us DATA — numbers we can use.

Expectation: A8.2

Learning Goal: "I can collect data by counting things in groups."

Materials: M1_sort_mats, M2_button_set, M3_tally_charts, M7_object_set

1. Minds On (8 mins): Tally Counts the Sort

Detective Dot sorted the buttons. Tally COUNTS them.

Bring children to the carpet. Detective Dot + Tally puppets. 'Friends — yesterday Detective Dot sorted. Today Tally COUNTS each group. We get DATA.'

2. Action (19 mins): Counting Stations

- Whole group: count 4 sorted button groups.
- Three stations.
- Stretch challenge at Station C: count more classmates and total.
- Discuss tally marks.
- Circulate to record formative evidence on the counting tracker.
- Two pairs share their counts of the same set.

3. Consolidation (8 mins): Tally's Count Brag

Discuss:

- What does DATA mean?
- How do tally marks help?
- Why bundle by 5s?
- What did you count today?

Exit: Detective Dot's refrain — class chants 3x: 'Tally's rule: ONE touch, ONE tally!' Each child writes one finding on a sticky.

Lesson 3: Picture Graphs

Goal: We can DISPLAY data with pictures. Each picture stands for one thing counted.

Expectation: A8.3

Learning Goal: "I can make a picture graph."

Materials: M3_tally_charts, M4_pictograph_template, M7_object_set, M9_data_chart

1. Minds On (8 mins): Tally Builds a Pictograph

Tally has tally counts. Now Tally turns them into PICTURES on a graph. Detective Dot opens: 'Pictures STACK — count from BACK!'

Bring children to the carpet. Detective Dot + Tally puppets. 'Friends — yesterday we COUNTED data with tallies. Today we DISPLAY data with PICTURES on a GRAPH.'

2. Action (19 mins): Pictograph Stations

- Whole group: build a pictograph from button counts.
- Three stations.
- Stretch challenge at Station C: pictograph with 4+ categories.
- Discuss alignment.
- Circulate to record formative evidence on the graphing tracker.
- Show two pretend pictographs of the same data — one aligned, one not.

3. Consolidation (8 mins): Tally's Graph Brag

Discuss:

- What does GRAPH mean?
- Why one picture per item?
- Why line them up?
- What does your graph show?

Exit: Detective Dot's refrain — class chants 3x: 'Pictures STACK — count from BACK!' Each child writes one finding on a sticky.

Lesson 4: Likelihood: Always, Sometimes, Never

Goal: Some events ALWAYS happen, some SOMETIMES happen, some NEVER happen — that's LIKELIHOOD.

Expectation: A8.4

Learning Goal: "I can tell if events ALWAYS, SOMETIMES, or NEVER happen."

Materials: M3_tally_charts, M4_pictograph_template, M8_class_chart, M9_data_chart

1. Minds On (8 mins): Likelihood word warm-up

Detective Dot says: 'Does the sun rise EVERY day? Does a fish FLY every day? Does it rain EVERY day?' Three different answers — that's LIKELIHOOD!

Bring children to the carpet. Detective Dot + Tally puppets. 'Friends — Detective Dot has THREE new words today. Listen for the difference.'

2. Action (18 mins): Likelihood corners

- WHOLE-GROUP MINI-LESSON (5 min): Detective Dot defines ALWAYS / SOMETIMES / NEVER with three benchmark events: 'we breathe air' (ALWAYS), 'it snows on Tuesday' (SOMETIMES), 'a...
- STATION A — ALWAYS HAPPENS: 4 children rotate to the ALWAYS corner.
- STATION B — SOMETIMES HAPPENS: 4 children rotate to the SOMETIMES corner.
- STATION C — NEVER HAPPENS: 4 children rotate to the NEVER corner.
- STRATEGY COMPARE: Two children share their SOMETIMES events.

3. Consolidation (8 mins): Detective Dot's Likelihood Brag

Discuss:

- What does ALWAYS mean?
- What does NEVER mean?
- What does SOMETIMES mean?
- What is the test?

Exit: Detective Dot's refrain — class chants 3x: 'ALWAYS — SOMETIMES — NEVER — that's LIKELIHOOD!' Each child writes one ALWAYS or NEVER event on a sticky.

Lesson 5: Data Detective Capstone

Goal: We can be DATA DETECTIVES — sort, count, display, read.

Expectation: A8.1, A8.2, A8.3, A8.4

Learning Goal: "I can be a Data Detective for a question I pick."

Materials: M1_sort_mats, M3_tally_charts, M4_pictograph_template, M5_question_cards

1. Minds On (8 mins): Detective Dot's Final Mission

Now you become a DATA DETECTIVE. Pick a class question. Sort, count, graph, describe. Detective Dot opens: 'SORT it! COUNT it! GRAPH it! READ it! That's how Data Detectives DO it!'

Bring children to the carpet. Detective Dot + Tally together. 'Friends — today YOU become DATA DETECTIVES. Pick a class question. Sort. Count. Graph. Describe.'

2. Action (18 mins): Detective Capstone

- WHOLE GROUP RECAP (5 min): four mini-stations on the carpet, each mirroring a prior day:
 - Station 1 (D1 — Sort): Detective Dot holds up 4 cards; class names ONE attribute and...
- Children build a pictograph from their survey.
- Stretch challenge: survey 12 classmates and graph.
- Children describe their graph: most, fewest, total.
- Whole-class share: 5-6 children share aloud.
- Two friends asked the same question but graphed it differently.

3. Consolidation (8 mins): Detective Dot's Closing Brag and Certificate Ceremony

Discuss:

- What's the most surprising survey you saw today?
- What's the most popular ___?
- Why are class graphs interesting?
- What other question would you investigate next?

Exit: Detective Dot's refrain — class chants 3x: 'SORT it!'

Name: _____

Date: _____

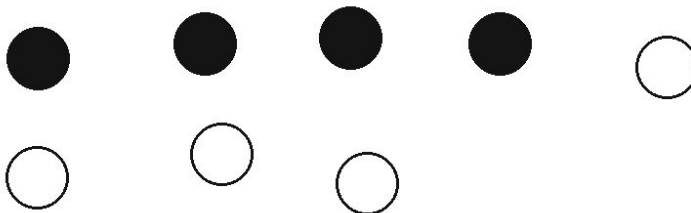
Detective Dot's Sort

Learning Goal: I can sort objects into groups.

Instructions for the Student:

1. Sort 12 picture-stickers into 3 colour groups.
2. Sort 8 shapes into BIG and SMALL.
3. Pick your own attribute.

Sort by COLOUR



DARK

LIGHT

--

--

Name: _____

Date: _____

Tally's Counting

Learning Goal: I can collect data by counting things in groups.

Instructions for the Student:

1. Count each sorted group.
2. Count things in your classroom by colour.
3. Survey 6 classmates.

Class Snack Tally

Question: _____

Item	Tally	Total
Apples		
Bananas		
Grapes		

Name: _____

Date: _____

Tally's Pictograph

Learning Goal: I can make a picture graph.

Instructions for the Student:

1. Use the counts to build a pictograph.
2. Build a pictograph from your survey (Lesson 2).
3. Pick 4 things to count and graph.

Sample Pictograph

Dogs



Cats



Fish



Name: _____

Date: _____

Always Sometimes Never

Learning Goal: I can tell if events ALWAYS, SOMETIMES, or NEVER happen.

Instructions for the Student:

1. Read each event.
2. Draw or write one event that ALWAYS happens at your school.
3. Write or draw an event that SOMETIMES happens.

Name: _____

Date: _____

My Detective Mission

Learning Goal: I can be a Data Detective for a question I pick.

Instructions for the Student:

1. Pick your question and 3-4 choices.
2. Survey 6 classmates.
3. Make a pictograph from your tallies.

My Question

Pick a question for your survey:

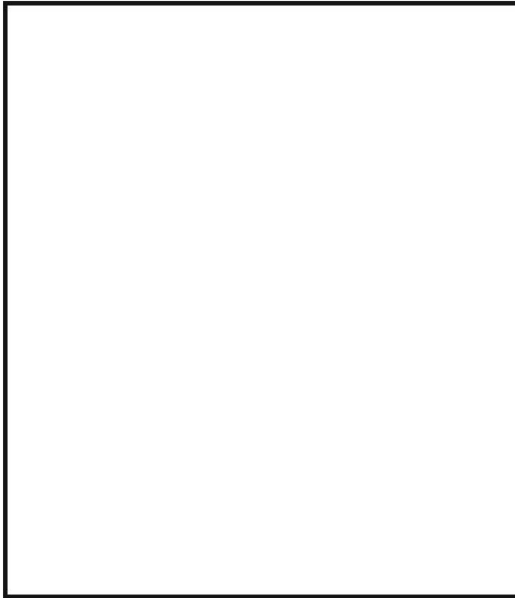
Sorting Mats

(M1_sort_mats)

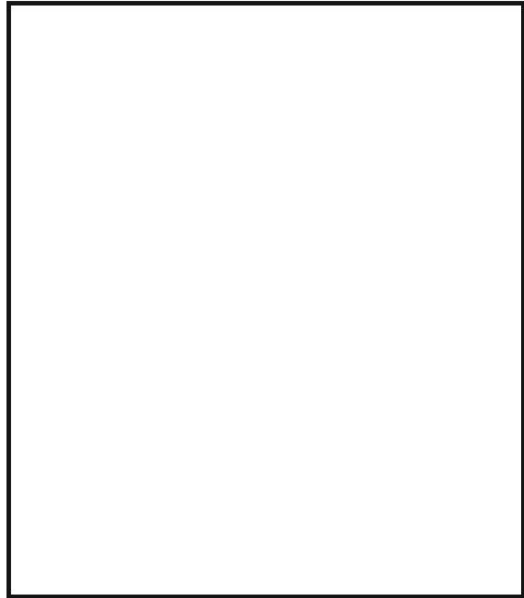
Letter-size sort mats with category circles for kindergarten sorting work.

My Sorting Mat

GROUP A



GROUP B



PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 mat per pair.

2. Laminate.

3. Hand out Lesson 1.

4. Reuse Lessons 2 and 5.

5. Use dry-erase.

USE IN CLASS

Quantity: 1 per pair

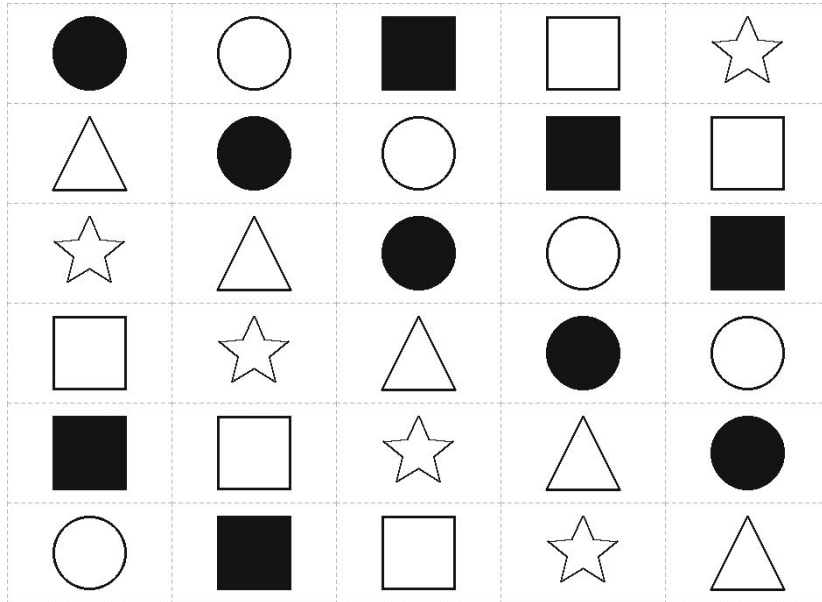
Lessons: 1, 2, 5

Prep: ~12 min

Paper Button Tokens

(M2_button_set)

60 paper button tokens in 4 colours for sorting and counting.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 5

Laminate: yes

USE IN CLASS

Quantity: 60 buttons, 4 colours

Lessons: 1, 2

Prep: ~25 min

TEACHER PREP

1. Print 5 pages on cardstock.
2. Colour-fill or use coloured cardstock.
3. Cut into 60 tokens.
4. Sort into bags.
5. Distribute Lessons 1-2.

Tally Chart Templates

(M3_tally_charts)

6 tally chart templates for recording counts.

Class Tally

Question: _____

Item	Tally	Total

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 6 per pair

Lessons: 2, 3, 4

Prep: ~12 min

TEACHER PREP

1. Print 1 page per pair.

2. Cut into 6 charts.

3. Laminate.

4. Hand out Lesson 2.

5. Reuse Lessons 3-4.

Pictograph Template

(M4_pictograph_template)

Letter-size pictograph template with rows.

Class Pictograph

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 per pair

Lessons: 3, 5

Prep: ~12 min

TEACHER PREP

1. Print 1 per pair.

2. Laminate.

3. Hand out Lesson 3.

4. Use dry-erase or stickers.

5. Reuse Lesson 5.

Class Question Cards

(M5_question_cards)

12 cards with class-data questions for survey practice.

Question Card

Do you like apples?

YES

NO

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 12 cards

Lessons: 1, 5

Prep: ~15 min

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Sort by topic.

5. Distribute Lessons 1 and 5.

Detective Capstone Template

(M6_capstone_template)

Letter-size template for Lesson 5 capstone investigation.

1. My QUESTION:
2. My DATA (tally):
3. My PICTOGRAPH:
4. What I FOUND:
5. My CONCLUSION:

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 1 per child

Lessons: 5

Prep: ~10 min

TEACHER PREP

1. Print 1 per child.

2. Hand out Lesson 5.

3. Children fill in.

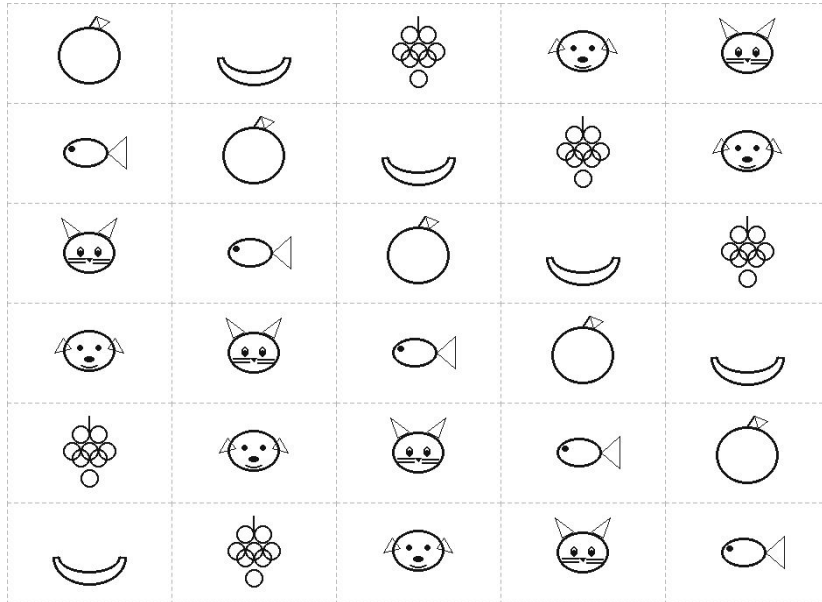
4. Collect for assessment.

5. Display.

Mixed Object Tokens

(M7_object_set)

60 paper tokens (shapes, animals, foods) for sorting variety.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 5

Laminate: yes

USE IN CLASS

Quantity: 60 mixed objects

Lessons: 1, 2, 5

Prep: ~25 min

TEACHER PREP

1. Print 5 pages on cardstock.

2. Laminate.

3. Cut into 60 tokens.

4. Sort into mixed bags.

5. Distribute Lessons 1, 2, 5.

Class Wall Chart

(M8_class_chart)

Letter-size chart for class-wide data display.

Class Chart

Use to record class data.

Name	YES	NO	?	Notes

PRINT SPECS

- Size: letter landscape
- Colour: bw
- Pages per set: 1
- Laminate: yes

USE IN CLASS

- Quantity: 1 chart
- Lessons: 2, 4, 5
- Prep: ~10 min

TEACHER PREP

1. Print and post on wall.
2. Laminate.
3. Use dry-erase.
4. Update with class data.
5. Reuse all week.

Data Anchor Chart

(M9_data_chart)

Anchor chart of the data process steps.

Data Chart	
ASK	Pick a question.
SORT	Group your data.
COUNT	How many in each?
SHOW	Make a graph.
TELL	What does it mean?

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 4

Prep: ~18 min

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall.

4. Reference all week.

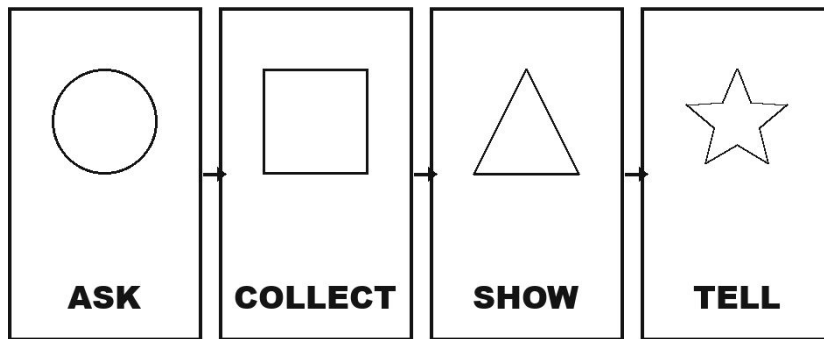
5. Roll for re-use.

Data Detectives Anchor Poster

(M10_detective_poster)

Anchor poster of Detective Dot and Tally.

Data Detectives!



Ask. Collect. Show. Tell.

PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 4
Laminate: yes

TEACHER PREP

1. Print 4 pages and tile.
2. Laminate.
3. Tape to wall.
4. Reference all week.
5. Roll for re-use.

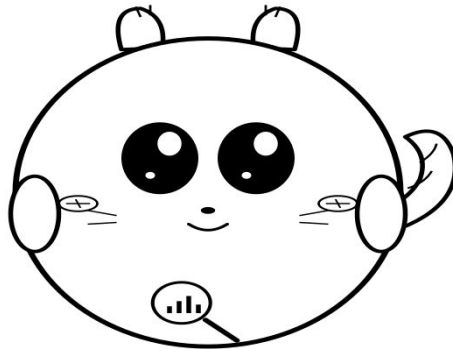
USE IN CLASS

Quantity: 1 poster
Lessons: 1, 2, 3, 4, 5
Prep: ~18 min

Detective Dot Character Puppet

(char_dot_puppet)

Printable Dot illustration on a popsicle stick.



Detective Dot the Squirrel

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

USE IN CLASS

Quantity: 1 puppet

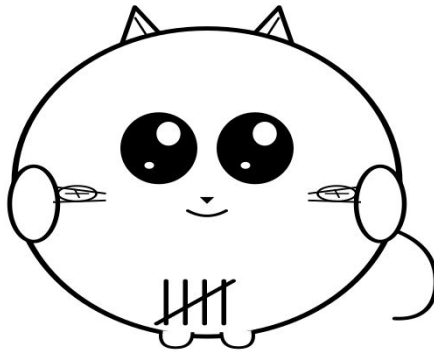
Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

Tally the Tally Cat Character

(char_tally_puppet)

Printable Tally illustration on a popsicle stick.



Tally the Tally Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
A8.1 sort objects or data using one attribute	Cannot sort objects into groups.	Sorts with prompts and supports.	Sorts independently by single attribute.	Re-sorts the same set by different attributes.
A8.2 collect and record data to answer questions of interest, and display it in concrete graphs or picture graphs	Cannot count or tally.	Counts with prompts and supports.	Counts and tallies 3+ groups accurately.	Bundles tallies in groups of 5.
A8.3 respond to data in concrete graphs and picture graphs by interpreting it and drawing conclusions	Cannot make a pictograph.	Makes pictograph with prompts and supports.	Makes pictograph with one picture per item.	Aligns rows for visual comparison; labels groups.
A8.4 describe how likely familiar events are to happen	Uses SOMETIMES as a catchall for every event.	Sorts ALWAYS / NEVER / SOMETIMES with prompts.	Independently sorts events into ALWAYS / SOMETIMES / NEVER.	Justifies SOMETIMES with 'not every day, not no day' and gives own examples.

Junior Data Detective Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Data Detectives unit and demonstrating the skills of a Junior Data Detective. Detective Dot and Tally congratulate the student.

By completing this unit, this student has demonstrated:

- Sorting objects into groups by attribute
- Counting and tallying data
- Making pictographs
- Reading and describing graphs
- Inventing a class data investigation

Marketplace Listing

A 5-day Ontario Kindergarten unit on data demonstrations (A8.1-A8.4) framed around 'Data Detectives' — Detective Dot the Sorting Squirrel and Tally the Tally Cat.

Price: \$9.99 CAD

Pages: ~32

Classroom time: 173 min

Teacher prep: ~90 min

Top tags: kindergarten, math, ontario-curriculum, data, sorting, graphing, canadian

What's included:

- 5 fully-scripted lesson plans with 50-minute durations and 250 minutes of total teaching time
 - 5 student worksheets aligned to each lesson
 - 1 mid-unit formative assessment (Data Check-In)
 - 10 printable teacher manipulatives plus 2 character puppets: sorting mats, paper button tokens (60), tally chart templates (6), pictograph template, class...
- 1 end-of-unit reflection sheet with 5 self-assessment prompts
- 4 class observation trackers: 1 diagnostic + 3 formative
- 1 four-row summative rubric covering A8.1-A8.4 across 4 levels
- 1 summative-administration guide tying the rubric to Lesson 5 capstone
- Junior Data Detective certificate (one per child)
- Detailed expanded walkthroughs covering common misconceptions
- Differentiation tracks built into every lesson